

WASP-36b

R-1869

Benchmark run · workflow W8-benchmark-deep · record deep-bench_WASP-36_191205 · created 2026-08-03T10:32:15+00:00

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2458822.9091 +/- 0.0035 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.166 +/- 0.049
Transit depth (Rp/Rs)^2	2.74 +/- 1.63 [%]
Orbital Inclination (inc)	87.42 +/- 2.83
Airmass coefficient 1 (a1)	115.79 +/- 134.76
Airmass coefficient 2 (a2)	-2.96 +/- 0.35
Scatter in the residuals of the lightcurve fit is	81.77 %
Best Comparison Star	None
Optimal Aperture	13.71
Optimal Annulus	14.27
Transit Duration (day)	0.092 +/- 0.01

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.166 ± 0.049 vs 0.1368 (deviation 0.38σ)
Duration fit vs published	0.092 d vs 0.0758 d (deviation 1.04σ)
Mid-time O–C	8.0 min
Depth SNR	2.2
Residual scatter	81.77 %

Light curve

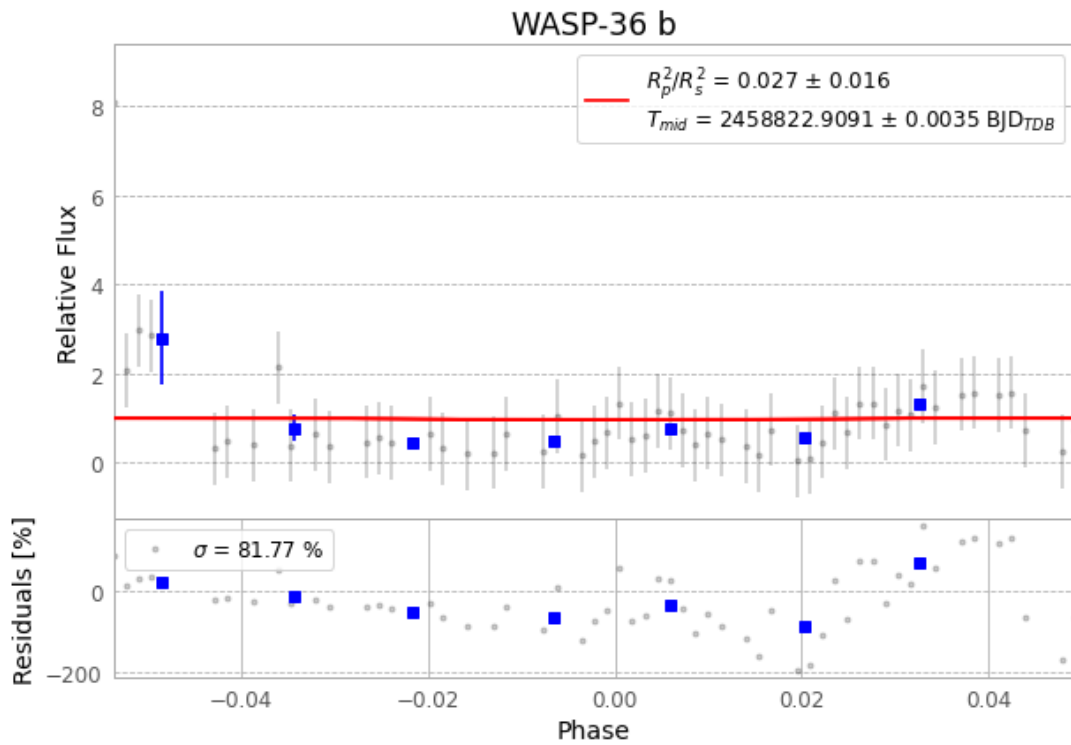


Figure 1 — FinalLightCurve_WASP-36 b_2019-12-05.png (SHA-256 a36ec97a906f7295...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [385, 350], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 5b775e2515d2d210...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b

Data provenance

77 FITS frames (51 MB), WASP-36191205074512.FITS → WASP-36191205113312.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-36-191205.log (1c980da75906...), FinalLightCurve_WASP-36 b_2019-12-05.png (a36ec97a906f...), FinalLightCurve_WASP-36 b_2019-12-05.csv (b27bea8a61a1...)

Compute resources

Wall time 270.5 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-03 10:32Z.